

Module 05: Action — Teaching a New Stitch Across the Circle

Learning Objectives

1. Describe teaching as an active inference process where the teacher adjusts their explanation based on the learner's prediction errors.
2. Analyze collaborative projects and group decisions as collective action — the circle acting as a unified agent.
3. Explain how action in the circle includes both physical action (stitching, demonstrating) and social action (helping, encouraging, deciding together).

Introduction

Rosa leans across the table. "Here, let me show you the bobble stitch. Watch my hands." She works a demonstration slowly, exaggerating each motion. James watches, tries to replicate it, fumbles. Rosa sees the fumble, pauses, and adjusts. "See this part? You are pulling up too soon. Wait until you have all five loops on the hook before you yarn over." James tries again. Closer this time.

This is **teaching** — and it is one of the purest examples of active inference in action. The teacher is not simply transmitting information; they are running a real-time inference loop. They predict what the learner needs, observe the result of their explanation, detect prediction errors (the learner's confusion or mistakes), and adjust their actions accordingly. The learner, simultaneously, is running their own inference loop — predicting what the stitch should look like, attempting it, encountering prediction error, and updating their model.

Key Concepts

1. Teaching as Active Inference

In Active Inference, action is the process by which an agent changes its environment to bring sensory input into alignment with its predictions. Teaching is a social form of this process: the teacher's goal is to change the learner's generative model so that it produces correct predictions about the stitch.

The teaching loop works like this:

1. **The teacher predicts** what the learner needs. Based on their model of the learner (how much they know, what they have struggled with before), the teacher generates a prediction about what explanation or demonstration will be most helpful.
2. **The teacher acts** — demonstrating, explaining, guiding hands.
3. **The teacher observes** the result — does the learner succeed? Do they look confused? Do they make the right motions?
4. **Prediction error** is generated if the result does not match the teacher's prediction. If the learner is still confused, the teacher's model of "what this learner needs" was wrong. The teacher updates.
5. **The teacher adjusts** — tries a different approach, slows down, uses an analogy, physically guides the learner's hands.

This loop continues until the learner's performance matches the teacher's prediction of success — or until both parties decide to take a break and try again later.

Notice that the teacher is not just modeling the stitch; they are modeling the learner. This is a **second-order inference**: the teacher is building a generative model of another agent's generative model. "James does not understand where to insert the hook, so I need to show him the insertion point more clearly." This metacognitive capacity — thinking about what another person is thinking — is what makes human teaching so powerful.

2. The Learner's Inference Loop

The learner is simultaneously running their own active inference process:

1. **The learner predicts** what the stitch should look like, based on the teacher's demonstration and their own prior knowledge.
2. **The learner acts** — attempts the stitch.
3. **Prediction error** arises when the result does not match expectations — the stitch looks wrong, feels wrong, or the yarn does something unexpected.
4. **The learner updates** — adjusting grip, tension, timing, insertion point.

The learner's prediction errors are the raw material of learning. Each error signals a gap between the learner's current generative model and the reality of the stitch. The teacher's role is to minimize the cost of these errors — to create a supportive environment where errors are informative rather than discouraging.

In a crochet circle, this supportive environment is built into the culture. Mistakes are expected. Frogging is normal. The collective ethos says: every expert was once a beginner, and every stitch you frog teaches you something.

3. Collaborative Projects as Collective Action

Beyond individual teaching moments, the circle engages in **collective action** — coordinated activity by the group as a whole. The most vivid example is the collaborative project.

When a circle decides to make a communal blanket — each member contributing squares that will be joined together — the group is acting as a collective agent. This requires:

- **Shared prediction:** a collective generative model of what the finished blanket will look like (colors, size, stitch pattern).
- **Coordinated action:** each member making squares that are compatible in size and gauge. This requires individual actions to be constrained by group-level predictions.
- **Distributed execution:** the work is divided among agents who each bring their own skill and style, yet must produce compatible outputs.
- **Integration:** someone (or the whole group) joins the squares together, resolving any mismatches between individual contributions.

This is active inference at the group level. The circle predicts a future state (the finished blanket), takes coordinated actions to bring that state about, encounters prediction errors (squares that do not quite match in size, colors that clash when placed together), and adjusts (remaking squares, rearranging the layout).

4. Social Action: Helping, Encouraging, Deciding

Not all action in a circle involves hook and yarn. **Social actions** — helping someone who is stuck, encouraging someone who wants to give up, deciding together what to work on next — are equally important forms of active inference in the social domain.

Helping is an action taken to reduce another agent's prediction error. When you see someone struggling and offer guidance, you are acting on your social generative model: "This person's distress is a prediction error in my model of a well-functioning circle. I can reduce this error by helping."

Encouraging is an action that adjusts another agent's precision. When you say "You are doing great, just keep going," you are increasing the learner's confidence in their own predictions — raising the precision of their model, which helps them persist through difficulty rather than giving up.

Group deciding is collective action in its most deliberate form. When the circle debates whether to attempt a lace pattern or stick with something simpler, they are collectively evaluating expected free energy — weighing the predicted difficulty (epistemic cost) against the predicted satisfaction (pragmatic value). The decision emerges from the interaction of multiple generative models, each contributing predictions and preferences.

Applications

- **Teaching Styles as Inference Strategies:** Different circle members teach differently — some demonstrate silently, some explain verbally, some guide hands physically. These are different strategies for reducing the learner's prediction error. A skilled teacher matches their strategy to the learner's model: a verbal learner benefits from explanation, a visual learner from demonstration, a kinesthetic learner from physical guidance. The best teachers switch strategies in real time based on the prediction errors they observe.

- **The Gift of a Finished Object:** When a circle member gives away a finished project — a hat for a friend's baby, a blanket for a charity drive — this is action that extends the circle's influence into the external world. The act of giving is an active state in the Markov blanket, carrying the circle's internal skill and care outward. The recipient encounters the object and, through it, the circle's generative model: the craft, the care, the hours of work.

Conclusion

Action in a crochet circle takes many forms — from the physical act of stitching to the social acts of teaching, helping, encouraging, and deciding together. Teaching is a particularly rich example of active inference, where teacher and learner run interleaved inference loops, each adjusting their actions based on the other's prediction errors. Collaborative projects reveal the circle acting as a collective agent, coordinating individual actions toward a shared predicted outcome. Social actions like helping and encouraging work on each other's generative models directly — adjusting beliefs, boosting precision, reducing isolation. In the next module, we explore how these actions accumulate into something deeper: learning that persists across time and generations.